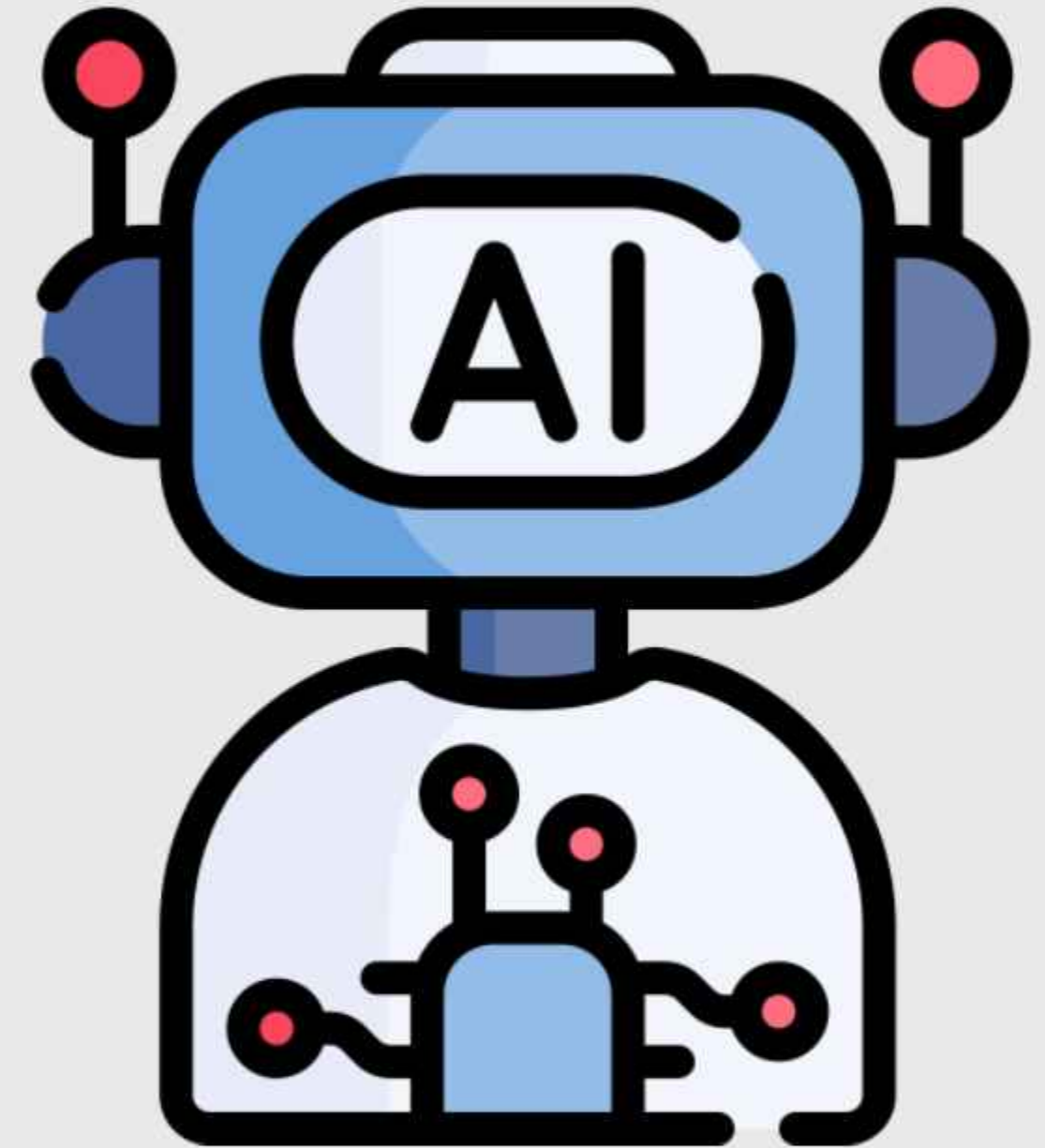
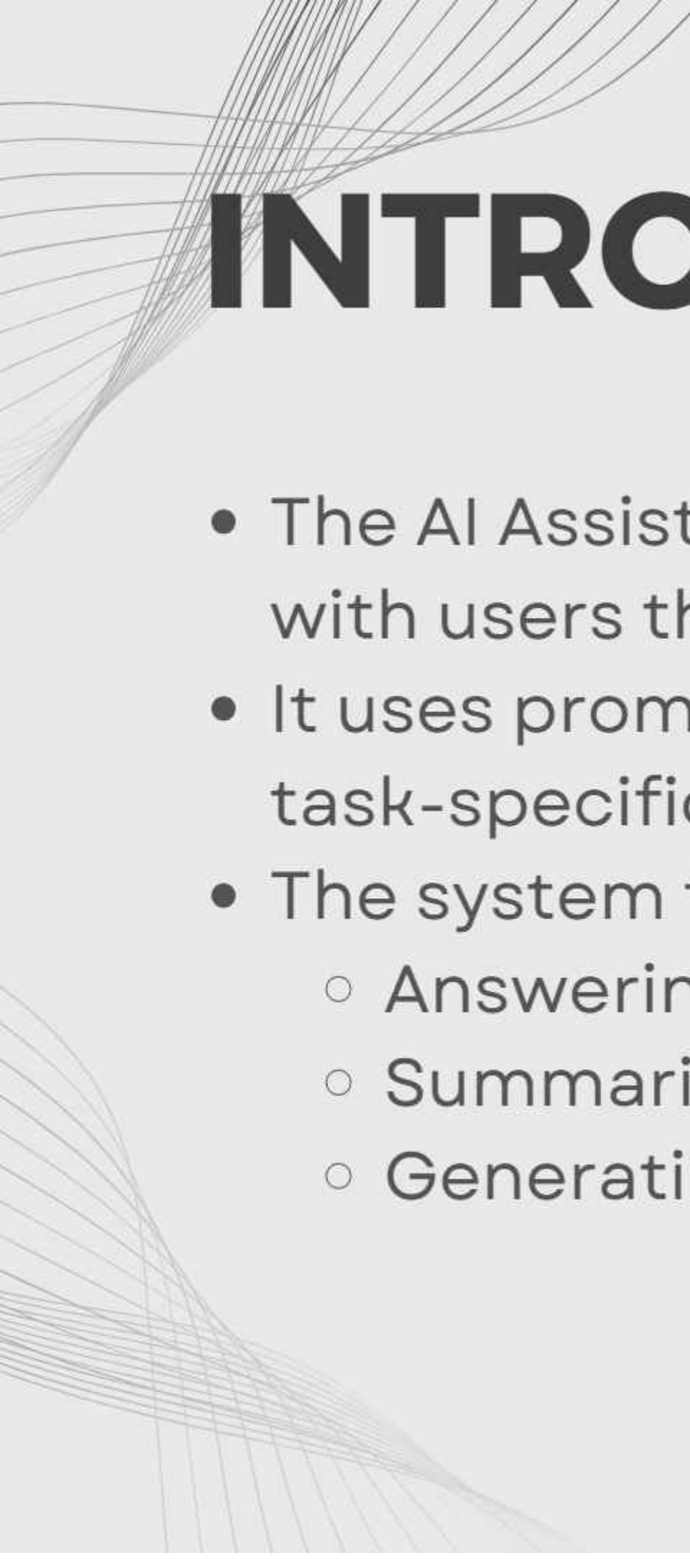


AI ASSISTANT – FUNCTIONALITY OVERVIEW

Prompt-Engineered Web
Application using
OpenRouter API

By: Sachin K





INTRODUCTION

- The AI Assistant is a web-based application designed to interact with users through natural language.
- It uses prompt engineering techniques to deliver accurate and task-specific responses.
- The system focuses on three primary capabilities:
 - Answering factual questions
 - Summarizing long text
 - Generating creative content

Function 1: Answer Questions

What it does: Responds to direct user queries by providing concise, factual, and accurate information.

Why it's useful: Saves time by eliminating the need to manually search through multiple sources.

How it works: The user inputs a question, and the assistant uses a structured prompt to ensure the response is brief and relevant.

Example:

- **Input:** "What is the largest planet in the solar system?"
- **Output:** "The largest planet in our solar system is Jupiter."

Applications:

- Quick educational lookups.
- Clarifying definitions and facts

Welcome to Your AI Assistant

What do you need help with today?

 Ask a question

Your prompt or text:

Type or paste your content here...

 Ask the Assistant

Function 2:

Summarize Text

What it does: Processes larger text blocks (articles, reports, or notes) and produces a shorter version containing key points.

Why it's useful: Helps users quickly understand the main ideas without reading the entire content.

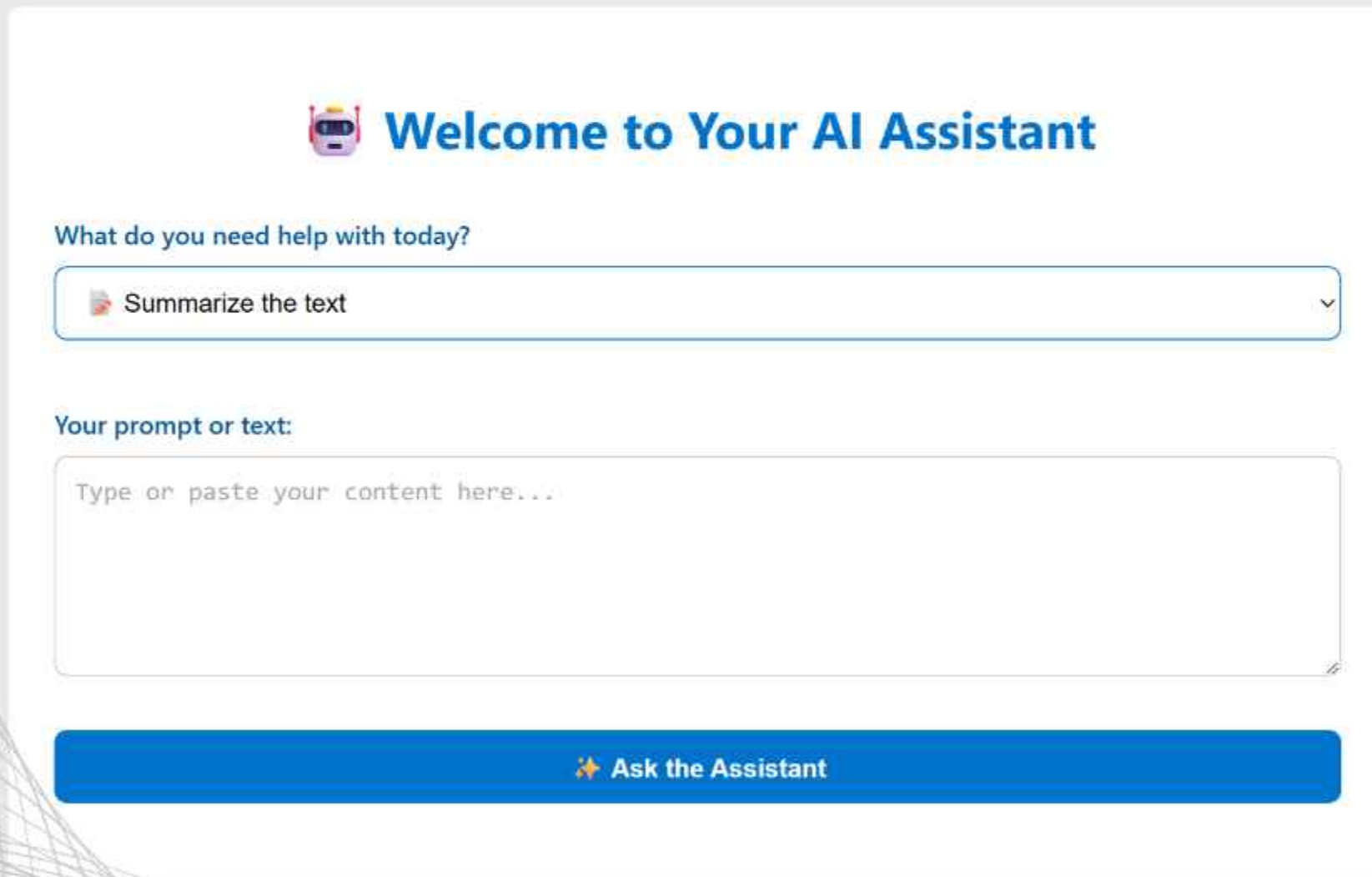
How it works: It uses pre-designed prompts to identify main ideas, remove redundancy, and generate a clear summary.

Example:

- Input: A 300-word article on climate policy
- Output: "The article discusses recent international climate agreements, their goals for reducing emissions, and challenges in implementation."

Applications:

- Academic research
- Preparing quick notes
- Reviewing reports



The screenshot shows a user interface for an AI assistant. At the top, there is a welcome message: "Welcome to Your AI Assistant" with a small robot icon. Below this, a prompt asks "What do you need help with today?". A dropdown menu is open, showing the selected option "Summarize the text". Below the dropdown, there is a text input field labeled "Your prompt or text:" with the placeholder text "Type or paste your content here...". At the bottom of the interface, there is a blue button with a star icon and the text "Ask the Assistant".

Function 3: Generate Creative Content



Welcome to Your AI Assistant

What do you need help with today?

Generate creative content

Your prompt or text:

Type or paste your content here...

Ask the Assistant

What it does: Creates original content such as short stories, poems, or brainstorming ideas based on user prompts.

Why it's useful: Supports creative writing, content generation, and idea exploration.

How it works: It applies a prompt designed to encourage imaginative responses while staying relevant to the input theme. **Example:**

- Input: "Write a poem about autumn leaves."
- Output: "Golden leaves drift through the air, whispering secrets everywhere..."

Applications:

- Creative projects
- Storytelling exercises
- Generating marketing or social media

FEEDBACK FEATURE

With the development of more precise ideas on elements, compounds and mixtures, scientists began to investigate how and why substances react. French chemist A. Lavoisier laid the foundation to the scientific investigation of matter by describing that substances react by following certain laws. These laws are called the laws of chemical combination. These eventually formed the basis of Dalton's Atomic Theory of Matter.

Feedback: yes

Prompt: In science, a law is a general statement that explains a large number of observations. Before being accepted, a law must be verified many times under many conditions. Laws are therefore considered the highest form of scientific knowledge and are generally thought to be inviolable. Scientific laws form the core of scientific knowledge. One scientific law that provides the foundation for understanding in chemistry is the law of conservation of matter. It states that in any given system that is closed to the transfer of matter (in and out), the amount of matter in the system stays constant. A concise way of expressing this law is to say that the amount of matter in a system is conserved.

Anything that has mass and takes up space.

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Feedback: yes

Prompt: what is the capital of India

Feedback: yes

Prompt: what is the capital of India

What it does: Allows users to indicate whether the generated response was helpful (Yes/No).

Why it's important:

- Enables tracking of performance.
- Provides data to refine prompts and improve response quality.

How it works:

Each interaction logs the feedback in a text file along with user input, the chosen function, and the generated output.

The background of the slide features a series of thin, grey, curved lines that flow from the top left towards the bottom right, creating a sense of movement and depth.

CONCLUSION

The AI Assistant demonstrates:

- Practical implementation of prompt engineering.
- Real-world applications for language models in education, content creation, and productivity.

- **Future directions:**

- Add user profiles, history tracking, and more functions.
- Provide analytics to study user needs and improve results.